

Atum V2 EVM/TRON Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Atum Labs on Atum V2 EVM/TRON. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated October 17, 2025, was delivered to Atum Labs, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Atum V2 EVM/TRON code provided by Atum Labs. The findings consist of 3 Informational findings.

Summary

Project Name	Atum V2 EVM/TRON
Repository	evm-contracts
Commit	ced6f29bea29...
Fix Commit	71206096773b...
Repository 2	tvm-contracts
Commit	a32d60551c65...
Fix Commit	ee83857255f9...
Audit Timeline	Oct 13th - Oct 17th, 2025
Methods	Manual Review, Formal Verification

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	0	0	0
Informational	3	1	2
Gas Optimizations	0	0	0
Total	3	1	2

6 Audit Scope

Atum V1 for EVM, TRON & Solana was previously audited by Cyfrin; the purpose of this audit is to evaluate the following V2 EVM/TRON changes:

1. New function call flow: deposit -> refund | release
    - the reserve function no longer exists; instead the reserve is done in the original deposit
    - both the deposit signature and the reserve signature are passed into the deposit function
2. Storage compression, including:
    - new witness strings and hashes
    - more parameters in each witness for reserve, release and refund
3. Support for Gnosis Safe smart contract wallets: `depositId` is now derived from `depositor`, `signature` and `permit.nonce` (previously from `depositor` and `signature` only)
4. Deadline parameters removed from the Release and Refund witnesses

5. New functions to batch each operation: `depositMany`, `releaseMany`, `refundMany`
6. Event structure changes to all events (`Deposit`, `DepositedMany`, `Released`, `ReleasedMany`, `Refunded`, `RefundedMany`, `Fulfilled`, `FulfilledMany`)
7. New `fulfillMany` function

The audit scope was limited to:

```
1-atum-evm/src/Escrow.sol
1-atum-evm/src/FulfillmentProxy.sol
2-atum-tvm/contracts/Escrow.sol
2-atum-tvm/contracts/FulfillmentProxy.sol
2-atum-tvm/contracts/TIP712.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Atum is a cross-chain stablecoin payments protocol that leverages an off-chain Request for Quote (RFQ) mechanism to facilitate seamless token transfers across different blockchains and stablecoin types. The protocol's design abstracts away blockchain complexity from the end users (Depositors), who simply need to sign a Permit2 message to initiate a payment request; they never need to interact directly with smart contracts or manage cross-chain messaging.

When a Depositor needs to pay in stablecoins across chains (e.g., deposit USDC on Ethereum but pay using DAI on Arbitrum), they publish an RFQ containing their signed Permit2 authorization, which multiple liquidity providers (Settlers) can bid on competitively. The source funds are locked in the Escrow contract while the winning Settler fulfills the swap on the destination chain, after which the escrowed funds are released to the Settler.

This architecture pushes all technical complexity, including liquidity management, cross-chain execution and blockchain interactions, to Settlers who compete to provide the best rates, while Depositors enjoy a simple, signature-based experience. This makes the protocol suitable for applications like cross-border payments, remittances and chain-agnostic stablecoin transfers.

9 Attestation

Cyfrin attests that it performed a security review of the Atum V2 EVM/TRON code provided by Atum Labs.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-10-17-cyfrin-atumv2-evm-tron-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Atum Labs has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=220324fc62640e1bcd3bde859f5c122e05305d28461a5ced9784f412601e2676
F=2025-10-17-cyfrin-atumv2-evm-tron-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.